Replication and reproduction

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Keep the artifacts. Count the cases. Separate new observations.

Explain replay, new-data evidence and case accounting without inflating distinctness or independence.

Keep one habit: Local definitions → eight-case replay ledger → four-record picture → mismatch probes → scoring/accounting audit → independent six-plus-four case.

MAIN LESSON · SECTION 1

Separate rerunning saved work from collecting new cases

> Count evaluation events, distinct cases and new evidence separately.

Learning goals

I can explain the **mechanism** and terminology of **replication** and reproduction.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve methods, outcome records and claim limits

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Methods results and limitations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Reading a research claim. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Methods results and limitations. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain **replication** and reproduction using the records and boundaries of this fictional investigation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of **replication** and reproduction

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Define replay, new-data evidence and the unit being counted

> Keep a successful rerun’s value without inflating population coverage.

Begin with eight questions counted twice

Imagine counting the same eight school questions twice. You now have two recorded runs, but not sixteen different questions. Replaying saved artifacts checks whether the recorded computation can be reproduced under the stated conditions. Collecting new eligible observations under a comparable protocol probes the question with new evidence. Define those terms in the project because naming conventions vary.

Neither a second run nor a new case ID automatically proves statistical independence.

Define reproduction and replication with local conventions

Reproducing a computation uses recorded artifacts; **replication** collects new **evidence** under a comparable question. Naming conventions vary, so define the terms in the project.

Complete artifact and case-accounting contract

Term or rule	Complete classroom convention
Replay/reproduce computation	Same saved records and stated analysis procedure/environment

Term or rule	Complete classroom convention
New-data replication	Newly collected eligible cases under a comparable question/protocol; here only supplied fictional new-data fixtures
Main accounting	Original eight IDs Q1–Q8 plus their replay =16 evaluation events, 8 distinct cases, 0 newly added cases
Record identity	Use case IDs, not whether two correctness values happen to be equal
Picture outcome meaning	Binary1=correct, 0=incorrect; correctness flags, not class labels
Picture original/replay	OLD1–OLD4 flags [1, 0, 1, 1]; same artifact/analysis gives 3/4=75%
Picture new-data fixture	NEW1–NEW4 flags [1, 1, 0, 0]; same scoring gives 2/4=50%
Independence	A collection-design assumption requiring scrutiny; distinct IDs alone do not establish it
Evidence status	No actual partner rerun, new collection or model study performed; all records are supplied practice fixtures

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace two runs without doubling the distinct case count

> Replaying evidence does not make it a new sample.

Original same-artifact versus new-data rule

Here **replay** means same artifacts/procedure; new-data **replication** means newly collected eligible cases under the same plan. Preserve environment, data versions and raw results.

Complete evaluation-event and case ledger

Main activity	Evaluation events added	Distinct case IDs after activity	What is learned
Original Q1–Q8 evaluated	8	8	Recorded outcome on those cases
Partner scenario reruns Q1–Q8	8	8	Same-score replay checks computational consistency
Main total	16	8	No new-case coverage added by replay
Claim of16 independent cases	Counts repeated events as new units	Unsupported	Neither distinctness nor independence doubled
No actual partner in local code	Only supplied IDs/replay calculation	No human outcome asserted	Record actual collaborator replay only if performed

State assumptions before counting evidence

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

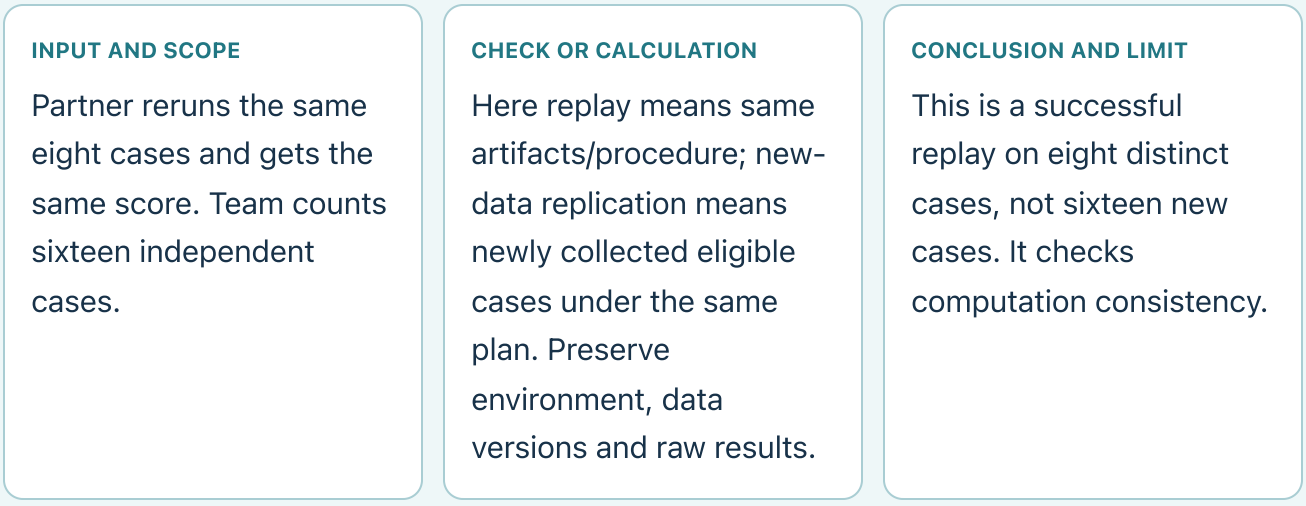
If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Replication and reproduction — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

Predict a mismatched rerun on unchanged saved records

> Investigate data, analysis and environment before accepting the mismatch.

Current task and response

- Partner reruns the same eight cases and gets the same score. Team counts sixteen independent cases. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict original records unexpectedly scored2/4

- Change: The rerun of the original records **output**s 2/4.

▼ Check after predicting

Prediction: Investigate code, version, **input** or scoring differences.

Reason: The same fixed records should give the same deterministic count.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

Read replay consistency and new-data variation as different questions

> Preserve each run and the artifact/protocol that produced it.

Main eight-case replay and checked conclusion

Partner reruns the same eight cases and gets the same score. Team counts sixteen independent cases.

This is a successful **replay** on eight distinct cases, not sixteen new cases. It checks computation consistency.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate correctness records and full arithmetic

Same records versus new experiment. A team first reruns a supplied analysis, then collects new records with the same protocol.

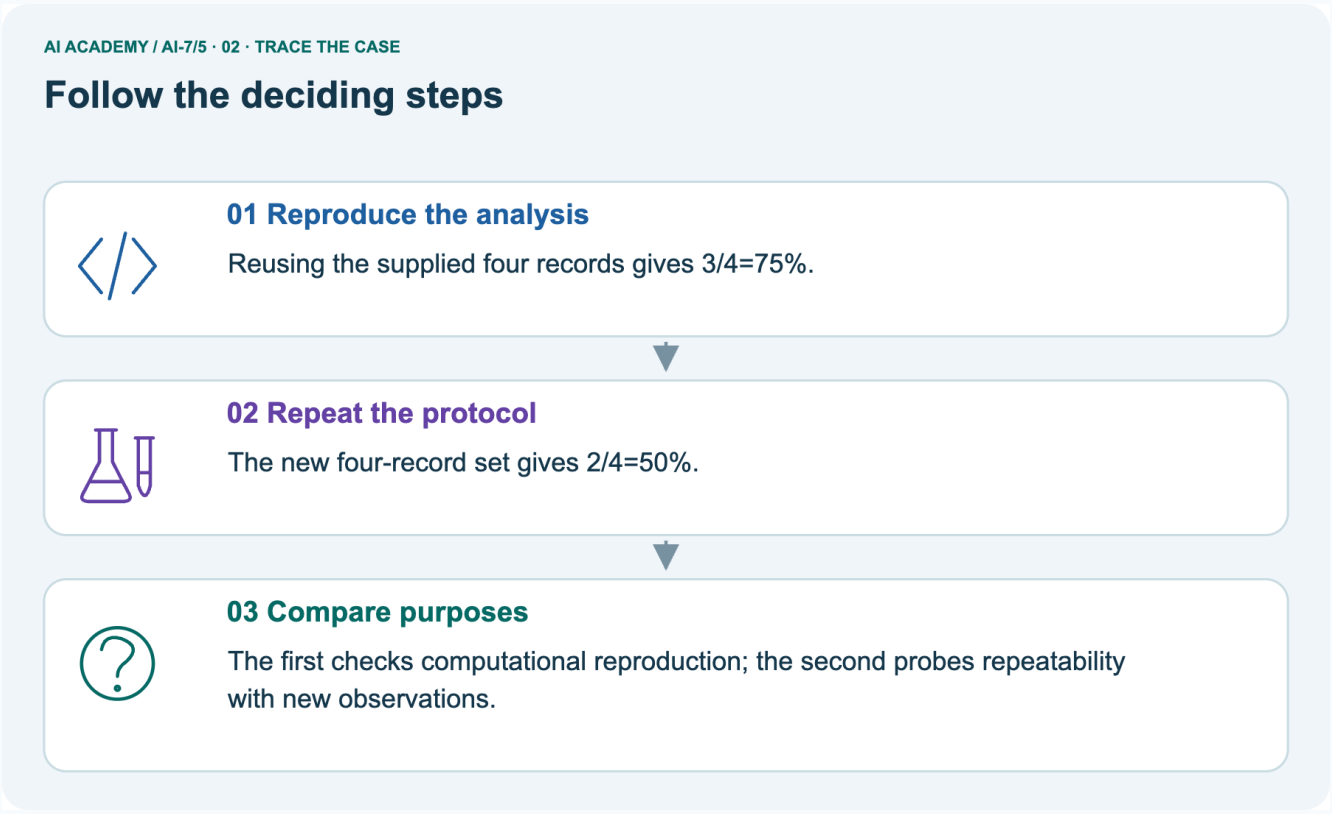
Original recorded correctness flags: [1,0,1,1]. Analysis counts correct cases. New independent set: [1,1,0,0]. Binary entries mean 1=correct and 0=incorrect.

The picture is a separate four-record example. Original correctness flags [1, 0, 1, 1] contain three correct outcomes, so the stated count/total analysis gives 75%. Replaying those same four records with the same deterministic analysis also gives 75%. New supplied records [1, 1, 0, 0] give 50% under that rule. A different new-sample score can arise without a code defect.

But a reported rerun of the original artifact giving 2/4 conflicts with the stated computation: inspect data/version changes, analysis, settings and transcription. The added code demonstrates one deliberately changed analysis as a possible mechanism, not the proven cause of the picture's hypothetical mismatch. The picture assumes newly independent observations; distinct IDs in a fixture do not verify that assumption.

Separate picture activity	Correctness flags	Score	Case accounting
Original saved analysis	1, 0, 1, 1	3/4=75%	4 original IDs
Replay same artifact/procedure	1, 0, 1, 1	3/4=75%	4 more events, 0 new IDs
New-data supplied fixture	1, 1, 0, 0	2/4=50%	4 new IDs
Across all three runs	12 evaluation events	Keep scores separate	8 distinct IDs, 4 newly added
Changed original rerun claims 2/4	Does not match supplied count rule	Investigate mismatch	Do not relabel as new-sample variability

Trace saved analysis, new-data protocol and purposes



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Reproduce the analysis	Reusing the supplied four records gives 3/4=75%.
2. Repeat the protocol	The new four-record set gives 2/4=50%.
3. Compare purposes	The first checks computational reproduction; the second probes repeatability with new observations.

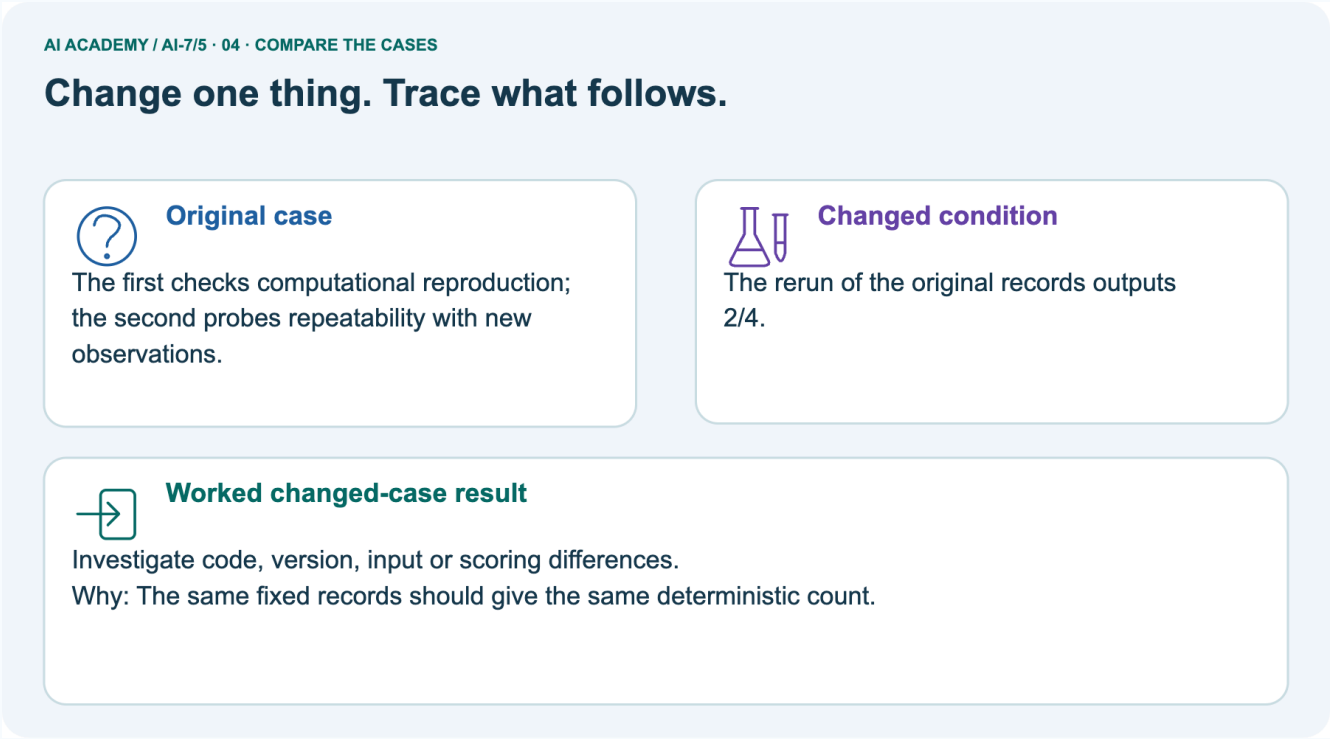
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Reusing data and collecting new data answer different questions. A different new-sample result can arise without a code error; inspect both protocols and variability.

Predict mismatched original rerun

- Change one thing: The rerun of the original records **output**s 2/4. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain small-sample and artifact-consistency limits

- Four-case samples are illustrative; neither result alone establishes broad reliability.
- Read the labelled picture: Calculate the scores from the original and new four-record sets.

  Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?

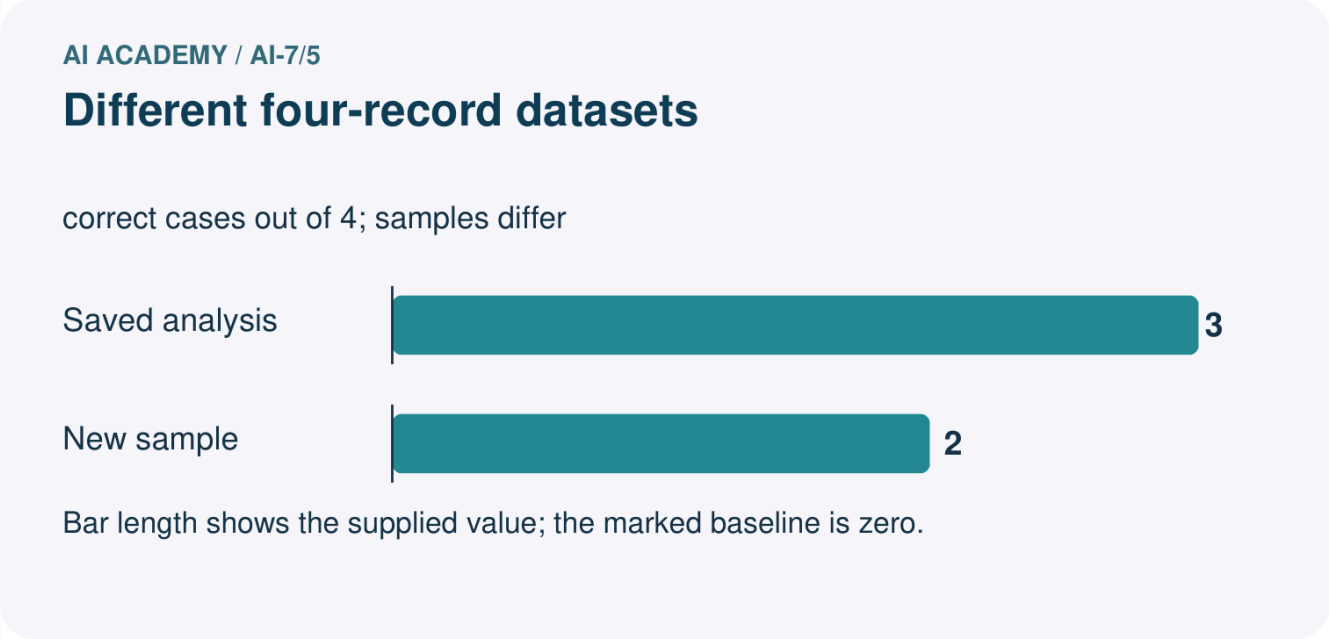
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Replication and reproduction

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/5

Replication and reproduction

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Reproduce the analysis

Reusing the supplied four records gives 3/4=75%.



2. Repeat the protocol

The new four-record set gives 2/4=50%.



3. Compare purposes

The first checks computational reproduction; the second probes repeatability with new observations.

CHANGE ONE THING

The rerun of the original records outputs 2/4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Audit inflated sample counts and overinterpreted mismatches

> Keep identity, protocol comparability and independence distinct.

Core reproduction qualification and failure analysis

Exact **replay** can preserve the original design flaws; new data can change conditions and needs its own record.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete counting and mismatch probes

Replay/replication probe	Expected interpretation
Same old cases rerun	Additional evaluation events, not new distinct cases
Different case IDs	Distinct identifiers, not automatic statistical independence
Different cases share same binary outcome	Still different cases; do not deduplicate by outcome value
Original deterministic records produce different score	Inspect artifact/code/environment/procedure mismatch
New eligible cases produce different score	May reflect sample variation or changed conditions; no automatic code-error conclusion
Same final aggregate score	Useful agreement but not proof every intermediate record/procedure matches
One known replay passes	Does not establish broader reliability or new-data replication
No collaborator actually reruns	Do not invent partner evidence

Replay/replication probe	Expected interpretation
New fixture supplied rather than collected	Label constructed new-data example, not completed collection study

Guided repair and explanation

- A report omits this qualification: Exact **replay** can preserve the original design flaws; new data can change conditions and needs its own record.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise explaining which evidence a second run adds

> Name original cases, replay cases and genuinely new cases.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 trace

Use workbook W2 for the connected example "**Replication** and reproduction". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for **Replication** and reproduction. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Calculate the scores from the original and new four-record sets. Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

Reserve the selected old-six/new-four accounting case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run the complete artifact replay and case-accounting investigation

> Inspect data/procedure identity alongside scores and distinct case counts.

Current eight-case sample-inflation investigation

Partner reruns the same eight cases and gets the same score. Team counts sixteen independent cases. Show the intermediate steps and give the conclusion with its scope.

Runnable supplied-record scoring and accounting

```
"""Fictional replay/new-data accounting; no partner or new experiment is claimed."""
from fractions import Fraction
from copy import deepcopy

def score(records):
 if type(records) is not list or not records:raise ValueError('nonempty case records
required')
 ids=[];correct=0
 for record in records:
  if type(record) is not dict or set(record)!={'case_id','correct'}:raise
ValueError('exact case ID and correctness fields required')
  if type(record['case_id']) is not str or not record['case_id'] or record['case_id'] in
ids:
   raise ValueError('unique nonempty within-run IDs required')
  if type(record['correct']) is not int or record['correct'] not in (0,1):raise
ValueError('exact binary outcome required')
  ids.append(record['case_id']);correct+=record['correct']
 return {'count':len(records),'correct':correct,'rate':Fraction(correct,len(records))}
def case_accounting(old_ids,replay_ids,new_ids):
 for ids in [old_ids,replay_ids,new_ids]:
  if type(ids) is not list or any(type(v) is not str or not v for v in ids) or
len(set(ids))!=len(ids):
   raise ValueError('unique symbolic IDs within each declared set required')
 if set(replay_ids)!=set(old_ids):raise ValueError('replay must use the same case IDs')
 if set(new_ids)&set(old_ids):raise ValueError('new-case set must not silently reuse old
IDs')
 return {'evaluation_events':len(old_ids)+len(replay_ids)+len(new_ids),
```

```
'distinct_case_ids':len(set(old_ids+replay_ids+new_ids)),'new_case_ids':len(new_ids),
        'statistical_independence':'not established by distinct IDs alone'}
main_ids=['Q'+str(i) for i in range(1,9)]
main=case_accounting(main_ids,list(main_ids),[])
assert main['evaluation_events']==16 and main['distinct_case_ids']==8 and
main['new_case_ids']==0
# Separate picture: supplied binary entries are correctness outcomes, not class labels.
original=[{'case_id':'OLD'+str(i+1),'correct':v} for i,v in enumerate([1,0,1,1])]
new=[{'case_id':'NEW'+str(i+1),'correct':v} for i,v in enumerate([1,1,0,0])]
before=deepcopy(original)
first=score(original);replayed=score(deepcopy(original));new_result=score(new)
assert first==replayed and first['rate']==Fraction(3,4) and
new_result['rate']==Fraction(1,2)
assert original==before
picture_counts=case_accounting([r['case_id'] for r in original],[r['case_id'] for r in
original],[r['case_id'] for r in new])
# A deliberately changed analysis is one possible mismatch mechanism, not a diagnosis.
buggy_rate=Fraction(sum(r['correct'] for r in original[:-1]),len(original))
assert buggy_rate==Fraction(1,2) and buggy_rate!=first['rate']
manifest={'analysis':'sum exact binary correctness / recorded count','data':'picture-
outcomes-v1',
          'procedure':'toy-score-v1','environment':'standard-library local calculation',
          'actual_partner_rerun':'not performed','new_data_collection':'not performed;
supplied fixture only'}
print('Main old-case replay accounting:',main)
print('Picture original/replay/new-data scores:',first,replayed,new_result,'case
accounting:',picture_counts)
print('Deliberately changed-analysis mismatch:',buggy_rate,'investigate
artifacts/code/settings before interpreting a mismatched replay')
print('Manifest and evidence status:',manifest)
print('A new-sample difference can arise without a code defect. These tiny supplied
samples prove no broad reliability, population independence or actual replication study.')
```

Expected outcomes and real-study limits

1. Predict main 16 events/8 distinct IDs/0 new cases, then original/replay 75% and separate new fixture 50%. The picture's three runs have 12 events, 8 distinct IDs and 4 newly added IDs. The code validates within-run IDs and exact binary outcomes, and rejects silently reused IDs in a declared new-case set.

   Its deliberately changed analysis omits the last original outcome but keeps denominator 4, producing an illustrative 2/4 mismatch; it is a possible bug demonstration, not a diagnosis of an actual rerun. The manifest explicitly says no actual partner or new-data collection performed. No population independence, model reliability or live replication is established.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: old six plus newly reserved four

> Separate repeated events from distinct and new-case evidence.

Complete selected independent task

Team repeats old six queries, then evaluates four newly reserved queries. How many distinct cases and which are new **evidence**?

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

Attempt this without hints

Team repeats old six queries, then evaluates four newly reserved queries. How many distinct cases and which are new **evidence**?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save artifact versions, procedure and each run separately

> Preserve raw fictional outcomes and honest collection/replay status.

Project task and artifact

Add a labelled **replication** and reproduction **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **replication** and reproduction.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual local scoring run or paper trace and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain what replay checks and what new observations add

> Avoid calling repeated cases independent new evidence.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **replication** and reproduction

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Replay	Repeating a computation with the recorded artifacts
Replication	A new evidence collection under a comparable declared question
Distinct unit	A separately eligible observation rather than a repeated copy

Use two terms to explain a step in this week’s example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: Exact **replay** can preserve the original design flaws; new data can change conditions and needs its own record.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

Explain which evidence supports your conclusion and which claim remains unproved.

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Replication and reproduction — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Partner reruns the same eight cases and gets the same score. Team counts sixteen independent cases.	Here replay means same artifacts/procedure; new-data replication means newly collected eligible cases under the same plan. Preserve environment, data versions and raw results.	This is a successful replay on eight distinct cases, not sixteen new cases. It checks computation consistency.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Same records versus new experiment. A team first reruns a supplied analysis, then collects new records with the same protocol.

By the end, I can	How I will show it
Explain the mechanism and terminology of replication and reproduction	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Same records versus new experiment. A team first reruns a supplied analysis, then collects new records with the same protocol.
- Which labelled step matches this detail? The first checks computational reproduction; the second probes repeatability with new **observation**s.
- 1. Compare purposes
- 2. Reproduce the analysis
- 3. Repeat the protocol
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare purposes The first checks computational reproduction; the second probes repeatability with new **observation**s.

Reusing data and collecting new data answer different questions. A different new-sample result can arise without a code error; inspect both protocols and variability.

Changed-case question and optional hint

- Changed-case practice: The rerun of the original records **output**s 2/4. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Investigate code, version, **input** or scoring differences.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Investigate code, version, **input** or scoring differences. The same fixed records should give the same deterministic count.

Repair and retain the record

Review Methods results and limitations (ai-7/4). Goal: Explain the **mechanism** and terminology of methods results and limitations.

Short practice: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories. Why cannot the result establish that B is best for outdoor farming?.

Return to this week and explain what you changed.

Review Reading a research claim (ai-7/2). Goal: Explain the **mechanism** and terminology of reading a research claim. Short practice: Calculate the baseline and new rates for 2/10 and 4/10.

Explain both twice the rate and the percentage-point change, and say why neither measures every learner's improvement.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

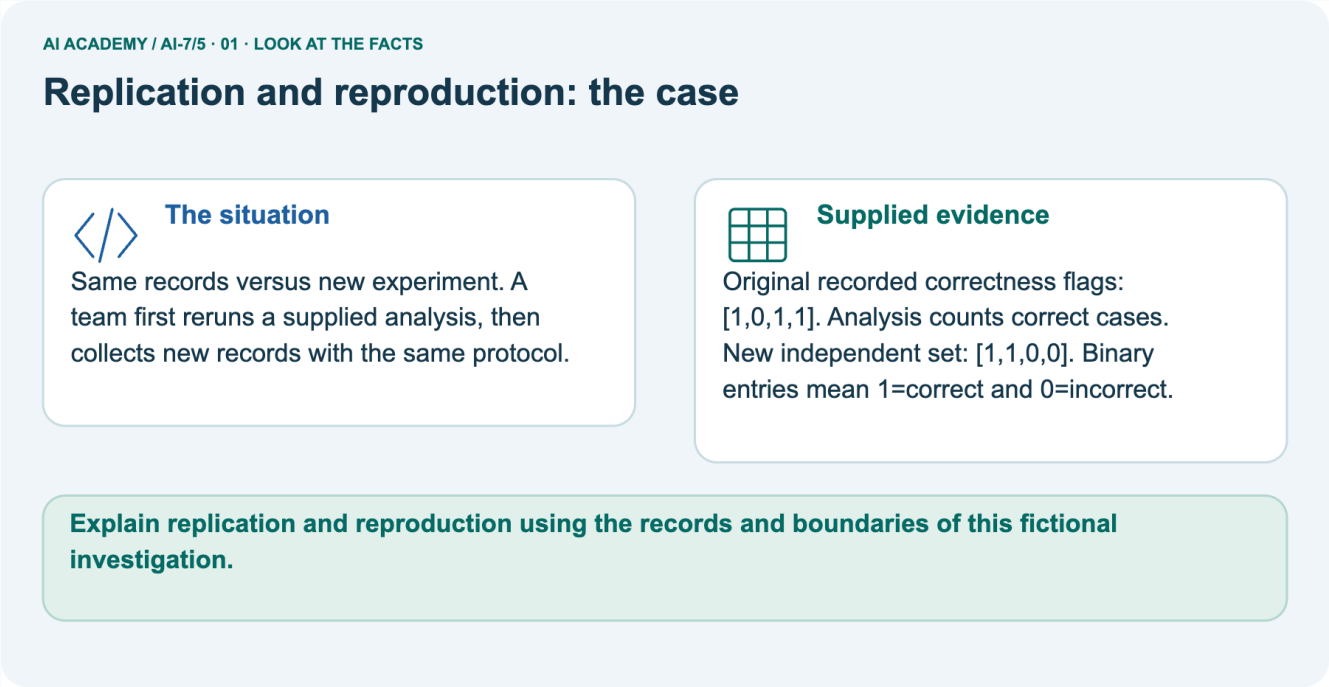
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Replication and reproduction: the case

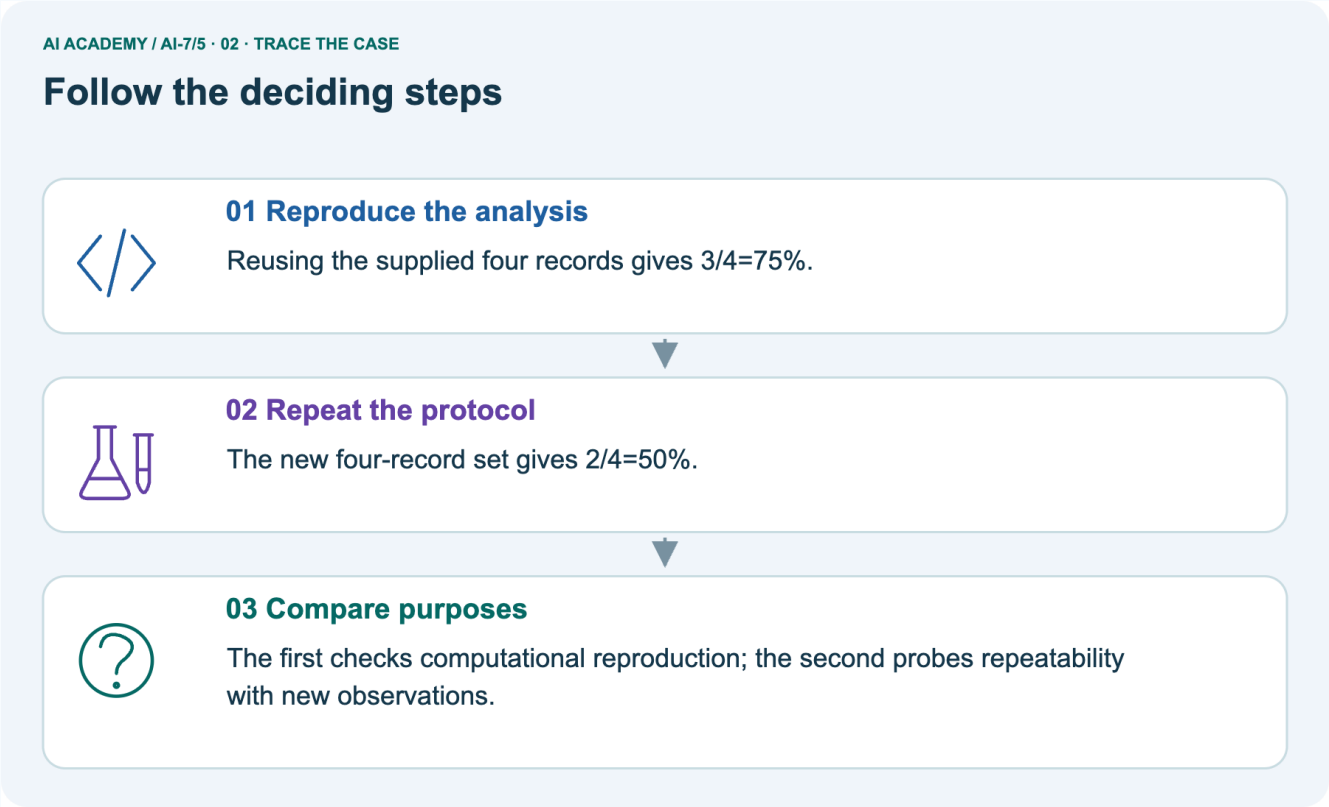


Follow the labelled picture with the matching explanation. Open the vector diagram

Same records versus new experiment. A team first reruns a supplied analysis, then collects new records with the same protocol. Supplied facts: Original recorded correctness flags: [1,0,1,1]. Analysis counts correct cases. New independent set: [1,1,0,0]. Binary entries mean 1=correct and 0=incorrect.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

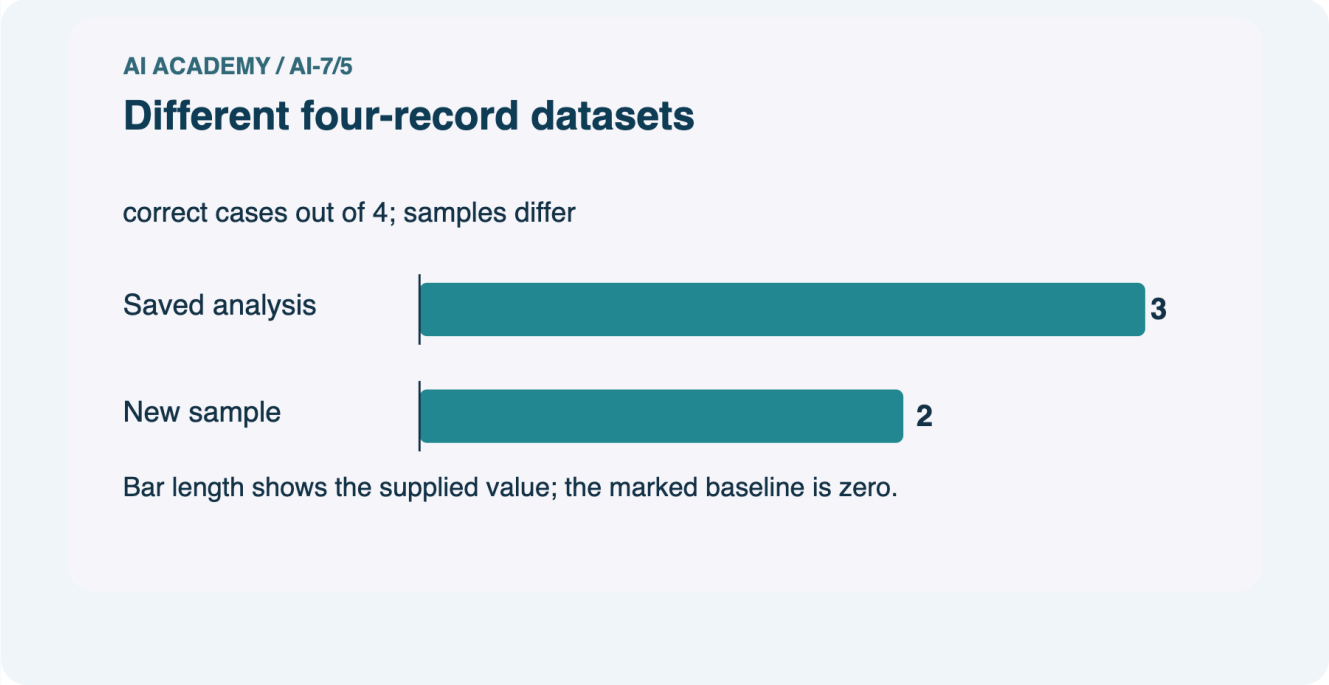


Follow the labelled picture with the matching explanation. Open the vector diagram

Reproduce the analysis: Reusing the supplied four records gives 3/4=75%. → Repeat the protocol: The new four-record set gives 2/4=50%. → Compare purposes: The first checks computational reproduction; the second probes repeatability with new **observation**s.

Point to each numbered stage and explain which recorded **input** it uses.

03. Different four-record datasets

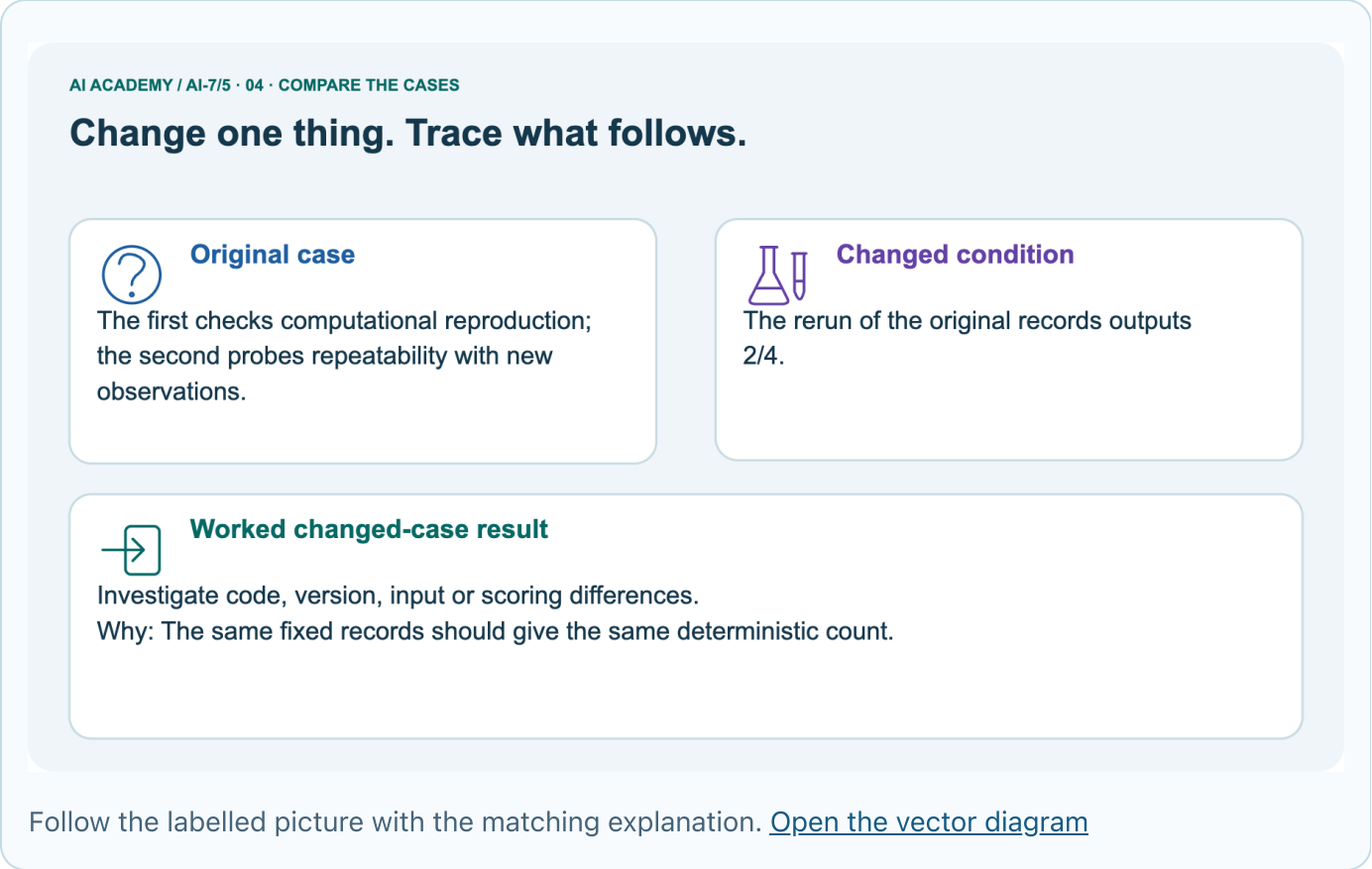


Follow the labelled picture with the matching explanation. Open the vector diagram

Different four-record datasets. Units: correct cases out of 4; samples differ. Saved analysis: 3. New sample: 2.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



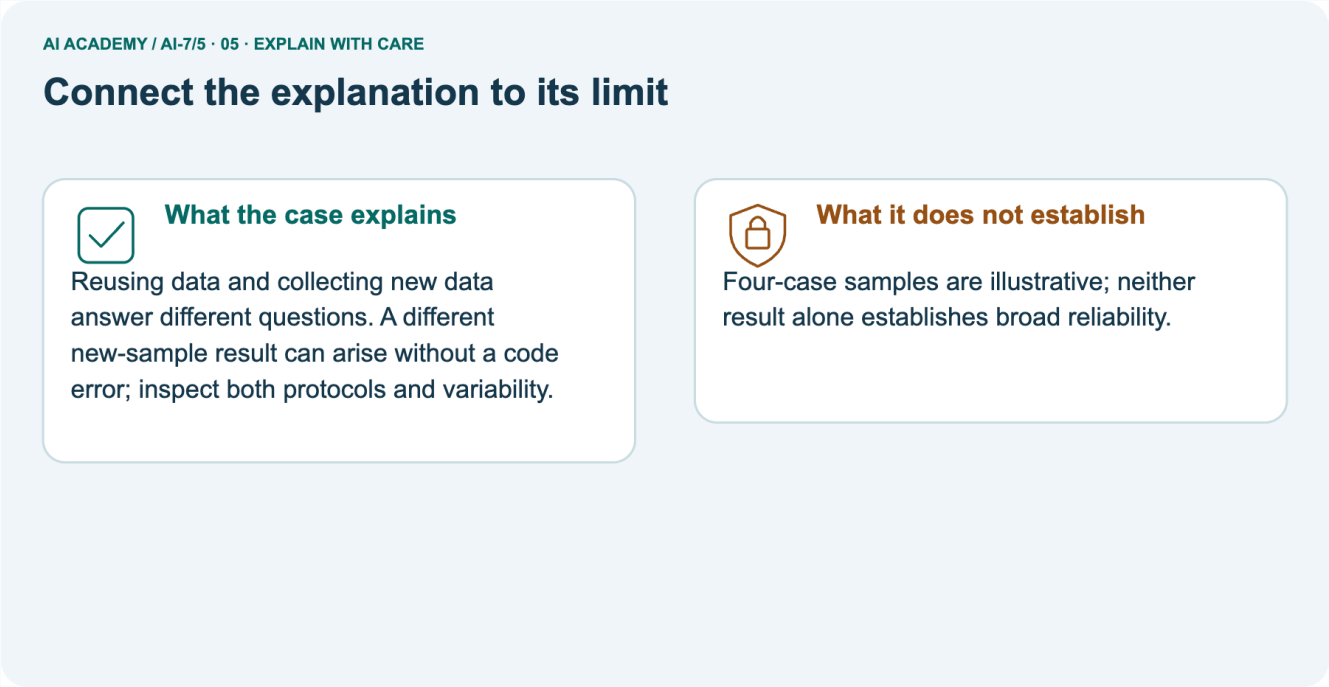
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The first checks computational reproduction; the second probes repeatability with new **observation**s. Changed condition: The rerun of the original records **output**s 2/4.

Expected result: Investigate code, version, **input** or scoring differences. Reason: The same fixed records should give the same deterministic count.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



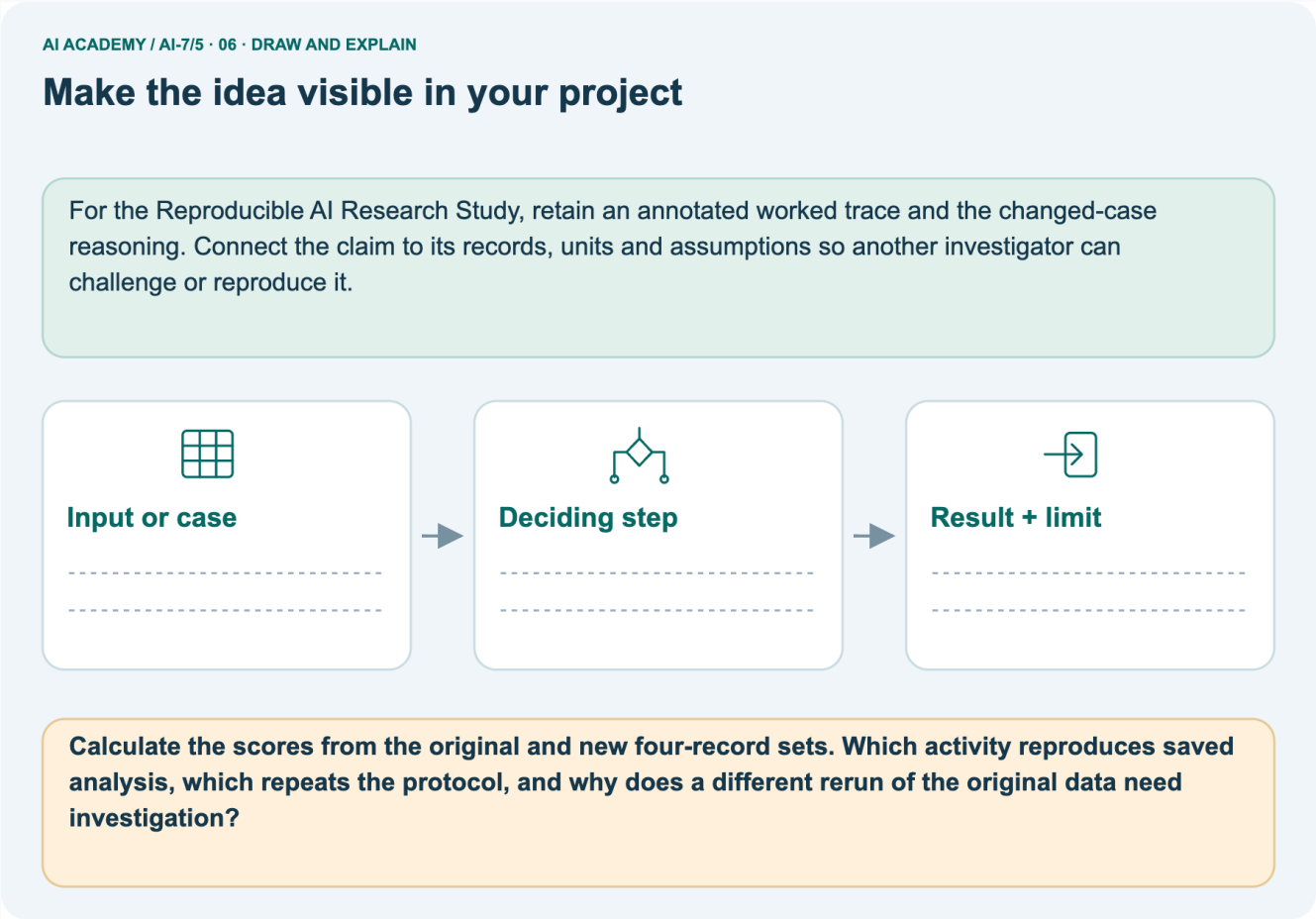
Follow the labelled picture with the matching explanation. Open the vector diagram

Reusing data and collecting new data answer different questions. A different new-sample result can arise without a code error; inspect both protocols and variability.

Limit: Four-case samples are illustrative; neither result alone establishes broad reliability.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the scores from the original and new four-record sets.

Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Replication and reproduction: the case

02. Follow the deciding steps

03. Different four-record datasets

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.